

Electric and Hybrid Vehicle Quick Reference

Fire Service Edition

INITIAL RESPONSE ACTIONS:

Conduct an Initial Scene Assessment to identify hazards.

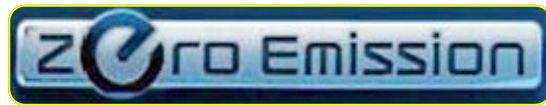
- Then: 1. **IDENTIFY** the drive system.
2. **IMMOBILIZE** the vehicle.
3. **DISABLE** high voltage & SRS.



Photo courtesy of Kevin Cooney

IDENTIFY

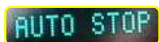
BADGES:



LABELS:



INSTRUMENTS:



COMPONENTS:

**Orange Cables
= High Voltage**



IMMOBILIZE

1. Chock wheels



2. Set Emergency **BRAKE**



3. Place vehicle in **PARK**



DISABLE

1. Ensure ignition is **OFF***



2. Disconnect 12V battery

***If vehicle is
equipped with
proximity key,
move it at
least 16 feet
away.**



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GENERAL WARNING AND CAUTIONS

- Never cut orange high voltage (HV), or yellow or blue medium (MV) voltage cabling. Never touch damaged or submerged HV or MV cables or components.
- Lack of engine noise in most hybrids and electrics does not mean that the vehicle is OFF. **Silent movement** or instant restart capability exists until vehicle is fully shut down.



SUBMERSION

- Vehicle chassis is safe to touch.
- High voltage (HV) system is **isolated** from chassis.
- Do not touch submerged HV cables or components.



FIRE

- High voltage (HV) battery fires may take much longer to extinguish than conventional fires.
- **Water** is the best extinguishing agent.
- Establish a water supply from hydrant or static source.
- **Smoke is toxic.**



POST INCIDENT

- Tow with a flatbed. Towing with drive wheels on ground may cause electrical fire.
- If high voltage (HV) battery is damaged, store vehicle at least 50 ft. from structures or vehicles.



- Monitor for signs of HV battery damage (inform tow operator).

For more detailed information and vehicle-specific alternate shutdown methods, see NFPA's **EV Emergency Field Guide**

DAMAGED HIGH VOLTAGE BATTERIES

- High voltage (HV) electrolyte leakage should be minimal, but is likely caustic and/or flammable *if* released.
- **Warning signs of hazardous damage:** sparks, smoke, Increasing temperature, gurgling/ bubbling sounds from HV battery.
- If any of these signs are observed, **ventilate the vehicle immediately**. The HV battery may be giving off harmful/ flammable gases and may become a delayed fire hazard.

